

Supplementary Material for VG-Mapping

I. INTRODUCTION

This supplementary document provides additional experimental analyses that could not be included in the main manuscript due to space limitations. Specifically, we investigate:

- Long-term stability under repeated scene revisits and updates.
- The influence of pose accuracy on reconstruction quality.

II. LONG-TERM STABILITY UNDER REPEATED UPDATES

Although the original VG-Scene benchmark is designed to evaluate map updates under semi-static scene changes, its sequences naturally support repeated-revisit evaluation. Each sequence contains a complete trajectory together with scene modifications. By repeatedly replaying the same sequence, the mapping system is exposed to recurring observations and repeated scene updates, enabling the evaluation of long-term map consistency under repeated revisits.

To evaluate the long-term stability of the proposed variation-aware density control (VDC) mechanism, we repeatedly replay the same semi-static sequence and record the rendering and geometric metrics after each revisit.

TABLE I
REPEATED-UPDATE EVALUATION OF VG-MAPPING.

Metric	Original	Repeat 1	Repeat 2	Repeat 3	Repeat 4
PSNR $\uparrow$	29.98	29.98	29.98	29.98	29.97
SSIM $\uparrow$	0.90	0.90	0.90	0.90	0.90
LPIPS $\downarrow$	0.14	0.14	0.14	0.14	0.15
Abs Rel $\downarrow$	0.041	0.041	0.042	0.043	0.043
$\delta_1 \uparrow$	0.94	0.94	0.94	0.93	0.93

As shown in Table I, the proposed method exhibits remarkable stability under repeated revisits. As the number of revisits increases from 0 to 4, all rendering and geometric metrics remain nearly unchanged. In particular, the PSNR changes by only 0.01 dB, while SSIM remains constant. Similarly, Abs Rel increases only slightly from 0.041 to 0.043 and δ_1 decreases from 0.94 to 0.93.

We attribute this behavior to the proposed VDC mechanism. By continuously removing outdated Gaussian primitives through variation-aware pruning while introducing new primitives only when necessary, the system effectively prevents the accumulation of obsolete scene representations. As a result, VG-Mapping maintains a consistent map representation even after multiple rounds of scene revisits and updates.

III. INFLUENCE OF POSE ACCURACY ON RECONSTRUCTION QUALITY

The proposed framework follows a tracking-mapping decoupled architecture, where camera poses are provided by

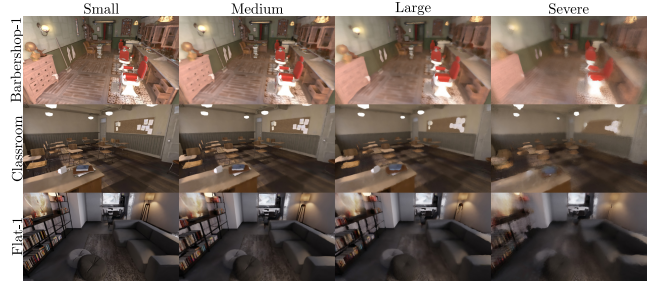


Fig. 1. Influence of pose perturbations on reconstruction quality.

an external visual-inertial odometry system. To investigate the influence of pose accuracy on reconstruction quality, we conduct a controlled perturbation study on three synthetic VG-Scene sequences (Classroom, Barbershop-1, and Flat-1).

Given a camera pose

$$\mathbf{T}_i = \begin{bmatrix} \mathbf{R}_i & \mathbf{t}_i \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad (1)$$

the perturbed pose is constructed as

$$\tilde{\mathbf{T}}_i = \begin{bmatrix} \Delta\mathbf{R}_i\mathbf{R}_i & \mathbf{t}_i + \Delta\mathbf{t}_i \\ \mathbf{0}^\top & 1 \end{bmatrix}. \quad (2)$$

The translational perturbation is sampled from

$$\Delta\mathbf{t}_i \sim \mathcal{N}(\mathbf{0}, \sigma_t^2 \mathbf{I}_3), \quad (3)$$

while the rotational perturbation is generated by independently perturbing the roll, pitch, and yaw angles:

$$\Delta\phi, \Delta\theta, \Delta\psi \sim \mathcal{N}(0, \sigma_r^2). \quad (4)$$

We consider four perturbation levels, namely Small, Medium, Large, and Severe, with $(\sigma_t, \sigma_r) = (0.5 \text{ cm}, 0.25^\circ), (1.25 \text{ cm}, 0.625^\circ), (2.5 \text{ cm}, 1.25^\circ), (5 \text{ cm}, 2.5^\circ)$, respectively.

TABLE II
INFLUENCE OF POSE PERTURBATIONS ON RECONSTRUCTION QUALITY.

Level	σ_t	σ_r	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
No Noise	0	0	32.76	0.94	0.05
Small	0.5 cm	0.25 $^\circ$	28.06	0.88	0.11
Medium	1.25 cm	0.625 $^\circ$	23.20	0.76	0.29
Large	2.5 cm	1.25 $^\circ$	21.24	0.729	0.393
Severe	5.0 cm	2.5 $^\circ$	19.46	0.706	0.473

Table II shows that reconstruction quality is highly dependent on pose accuracy, which is consistent with observations reported in other Gaussian-based mapping systems. Even under the Small perturbation level, the PSNR decreases from

32.76 dB to 28.06 dB, while LPIPS increases from 0.05 to 0.11. As the perturbation level increases, rendering quality degrades progressively.

Interestingly, despite the noticeable degradation in rendering quality, the proposed framework remains capable of correctly identifying and updating changed regions in the semi-static scene. As illustrated in Fig. 1, the latest scene state can still be successfully incorporated into the map even under moderate pose perturbations.

This observation suggests that pose errors primarily affect the fidelity of the reconstructed geometry and appearance, whereas the variation-aware update mechanism itself remains effective. Therefore, pose accuracy determines the upper bound of reconstruction quality, while the proposed VDC mechanism remains capable of detecting and updating semi-static scene changes.